



Medical Microbiology

Third stage- College of Dentistry

Lab. 8

Lactobacillus

Prepared by:

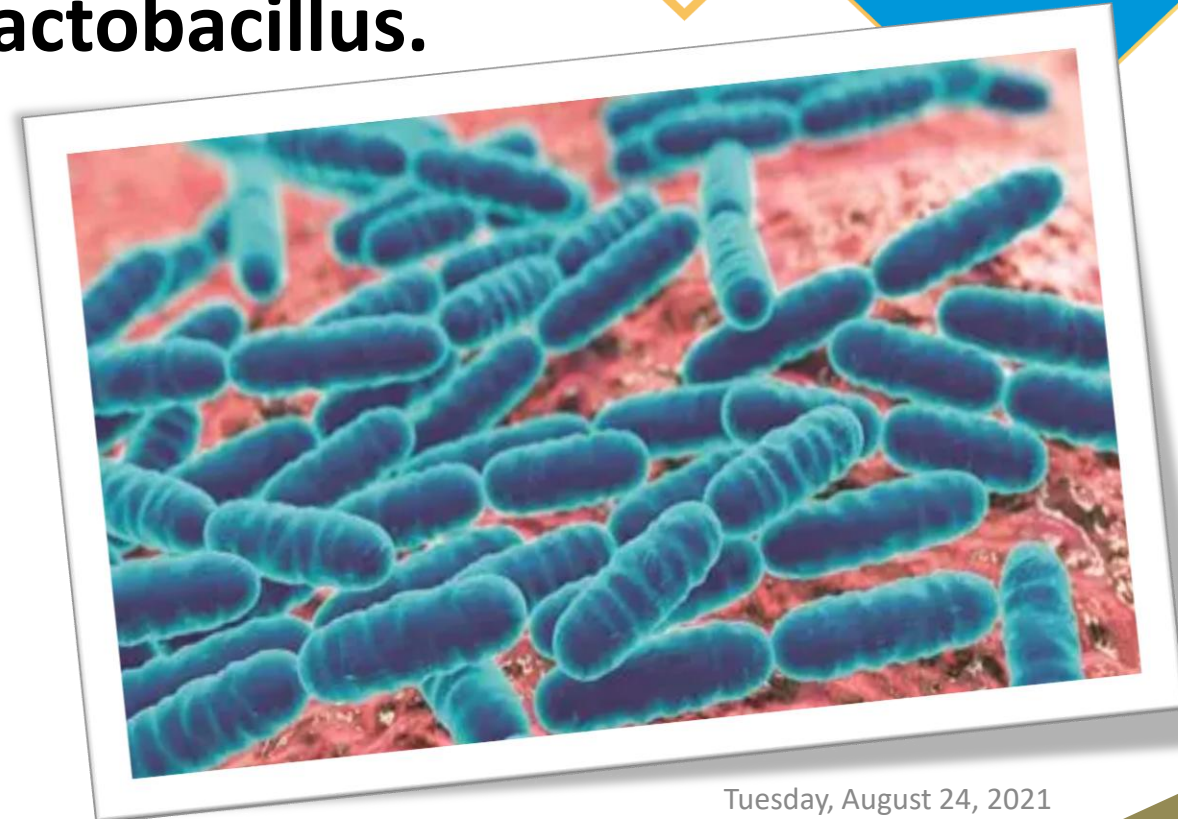
PhD. Dheyaa Naji Hamza

M.Sc. Zainab Majid Mohammed

Learning Objectives:

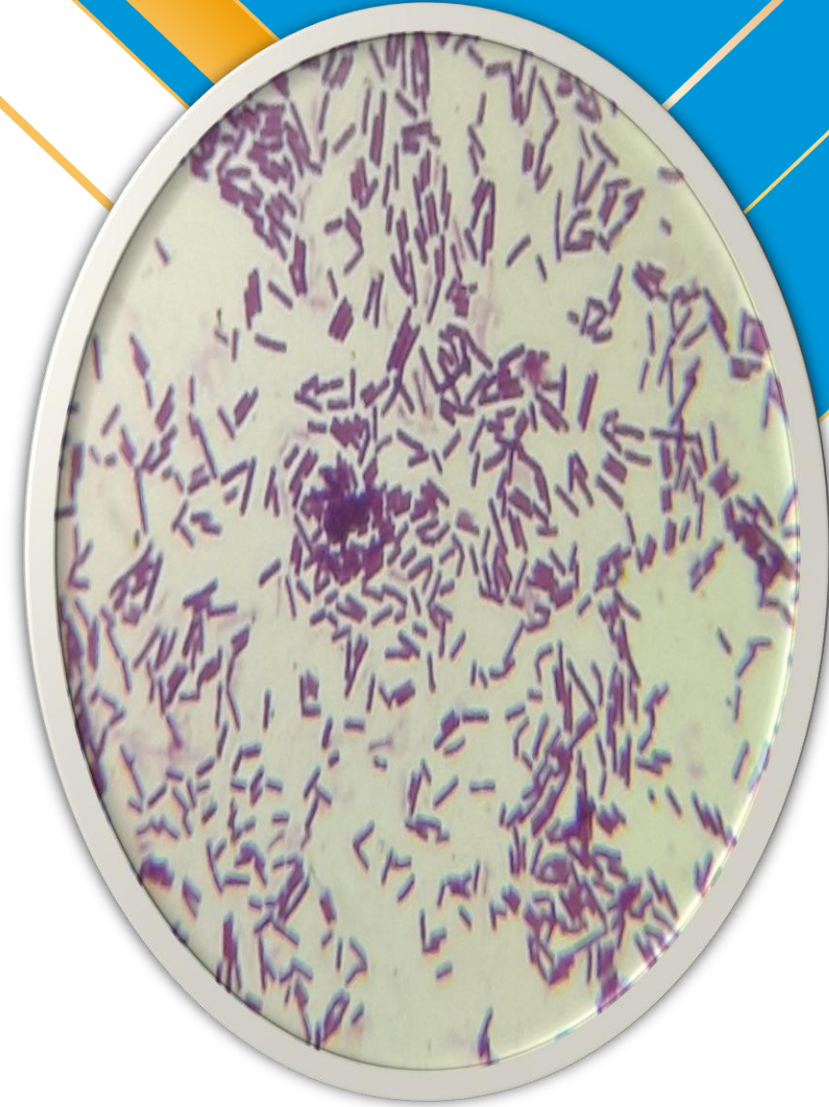
After this lab. You must be able to:

- 1-Describe Lactobacillus under microscope.**
- 2-Distinguish types Fermentation of Lactobacillus.**
- 3- know of Practical Applications of Lactobacillus.**



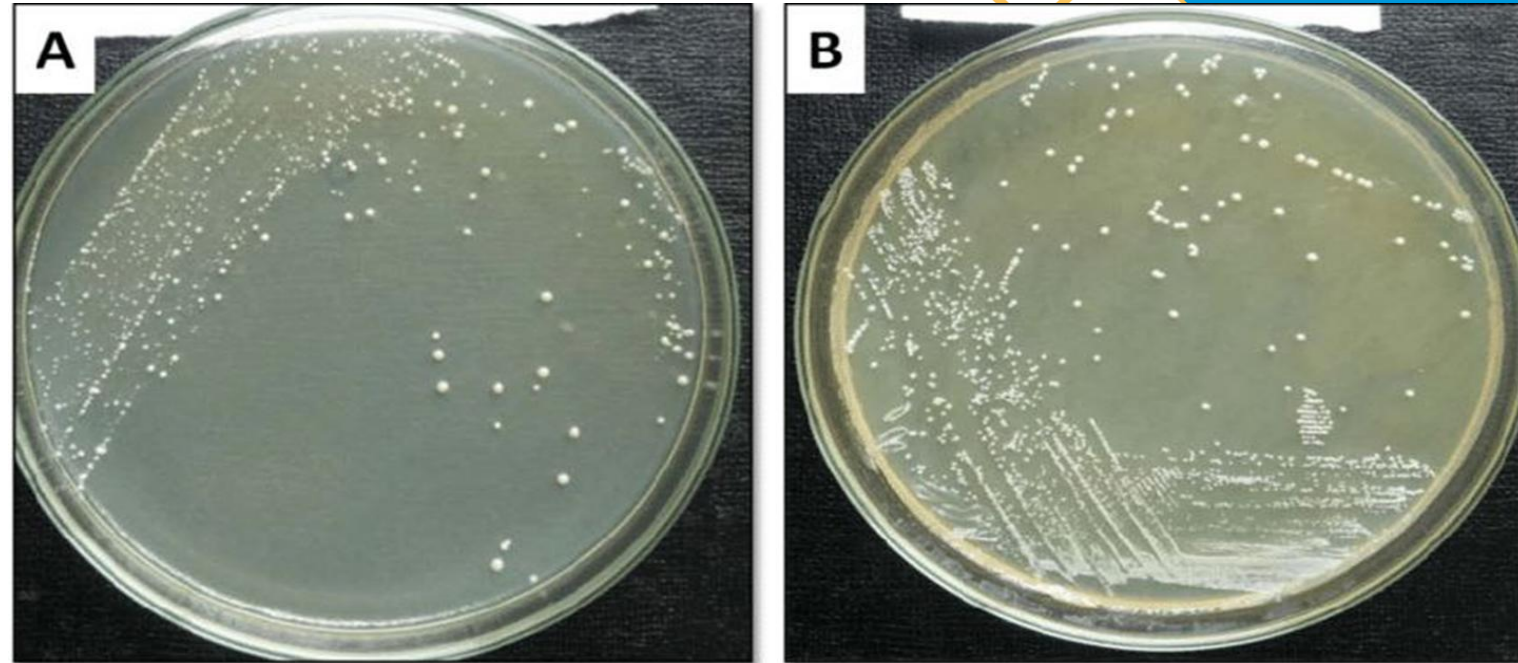
Morphology of *Lactobacillus*

- Gram positive
- Rod-shaped (bacilli)
- Facultative anaerobic or microaerophilic.
- Non-spore forming.
- Non-motile.
- Non- capsule.



Habitat: Commonly found in the gastrointestinal and genito-urinary tracts of mammals (normal flora/microbiota), and in fermented food and dairy products.

Culture Media: Use MRS Agar/Broth (Man Rogosa Sharpe) as it is selective and favorable for Lactobacillus growth.



Lactobacillus in MRS media

Metabolism (Fermentation):

They are a major group of Lactic Acid Bacteria (LAB), converting sugars into lactic acid via fermentation.

-**Homofermentative**: Produces mainly lactic acid, Ex: **L. acidophilus**

-**Heterofermentative**: Produces lactic acid, CO₂, and ethanol/acetic acid, Ex: **L. brevis**

Homolactic fermentation



Fermentation of cheese and yoghurt

Heterolactic fermentation

Lactic acid

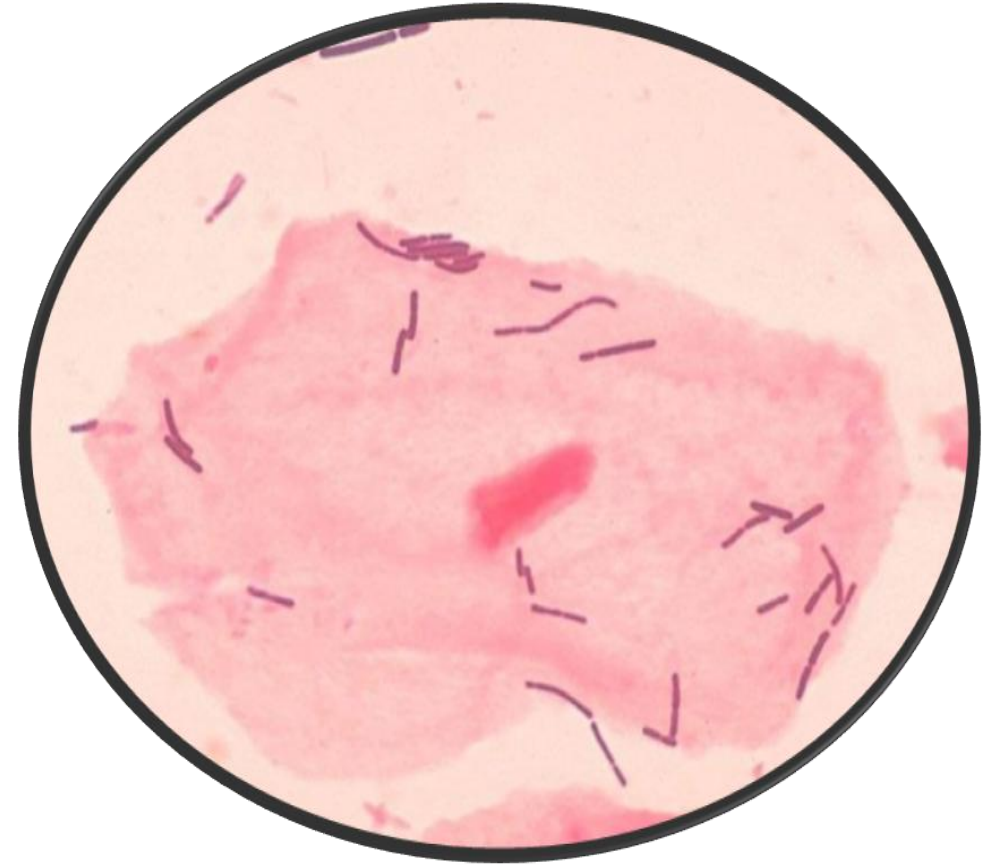
Acetic acid
CO₂



Spoilage of many food products

Practical Applications

- **Food Industry:** Used as starter cultures for producing yogurt, cheese, and pickles. Their lactic acid production is key for preservation and flavor.
- **Probiotics:** Many species are known probiotics (*L. rhamnosus*, *L. plantarum*), that benefit human gut health and immunity.



Biochemical Test of Lactobacillus spp.

Capsule- Negative

Catalase-Negative

Oxidase-Negative

Motility-Negative

Lactose-positive

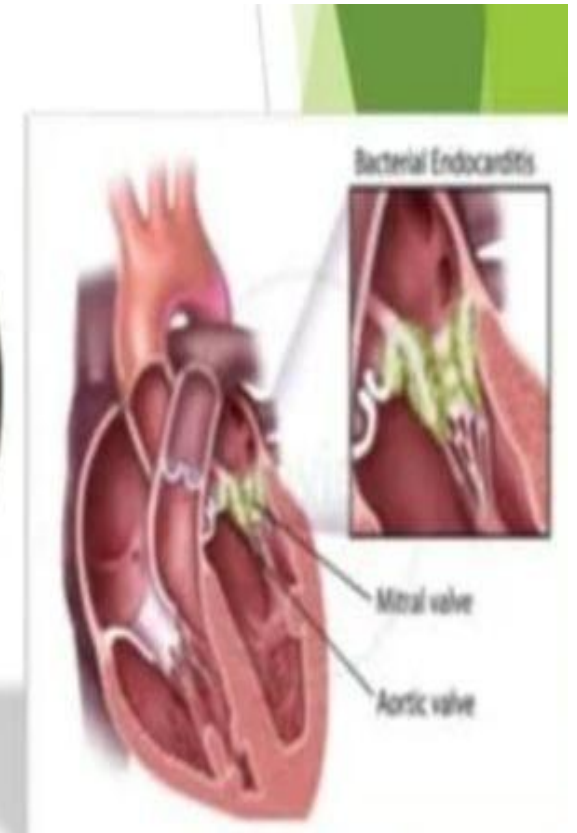
Galactose-positive

What are the virulence factors of Lactobacillus?

- The virulence factors mainly associated with lactobacilli consist in the ability of some strains to produce specific enzymes as **glycosidase** (are enzymes that break down complex carbohydrates and glycoproteins by hydrolyzing glycosidic bonds) and **arylamidase proteases** (is an enzyme that catalyzes the hydrolysis of an N-terminal amino acid from arylamides, or peptides and amides).
- It is involved in the breakdown of proteins and peptides and plays a significant role in soil [nitrogen mineralization](#).
- The activity of this enzyme can be influenced by factors like pH, metal ions, and microbial activity.

Pathogenicity

Most are non- pathogenic but some species can cause dental caries , atopic dermatitis and endocarditis.





Any Question?



Thank you